

SWEEPING SECULAR RESONANCES IN THE KUIPER BELT CAUSED BY DEPLETION OF THE SOLAR NEBULA

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ABSTRACT

We have investigated excitations of orbital eccentricities and inclinations of Kuiper belt objects (KBOs) caused by the sweeping secular resonances during the primitive solar nebula depletion. Since nebula gravitational potential rotates the longitudes of perihelia and ascending nodes, the nebula depletion leads to migration of secular resonances. In the outer (classical) Kuiper belt (the region beyond 42 AU), inclinations and eccentricities are respectively distributed up to 0.6 (radian) and 0.2, and their root mean squares are about 0.2 (radian) and 0.1. These large values are not explained by present planetary perturbations alone. We have investigated the sweeping secular resonances in the Kuiper belt with both direct orbital integration and the analytical method and have found that the sweeping secular resonances can account for the eccentricity and inclination in the outer belt. Inclinations of objects in the outer belt are excited to the observational level if the residual nebula with about 0.1% of the density in the minimum mass nebula model is depleted in a timescale of 10^7 – 10^8 yr. For inclination excitation, Jovian perturbations and nebula potential are the most important, and Neptunian perturbations do not play an important role during the residual nebula depletion, although Neptune with more than one-fifth of its present mass is needed for enough eccentricity excitation. If further observation of the KBOs at semimajor axis $\gtrsim 50$ AU confirms our model, it would give important clues about Neptune's formation and the depletion of the solar nebula.

Key words: celestial mechanics, stellar dynamics — Kuiper belt, Oort cloud — solar system: formation

1. INTRODUCTION

More than 280 small objects have been detected in the Kuiper belt region, and orbital elements of more than 200 objects are identified.¹ Figure 1 and Figure 2 show the distribution of eccentricities and inclinations of Kuiper belt objects (KBOs). In this paper, inclinations are expressed in units of radians with respect to the ecliptic plane. The orbital distribution of KBOs is roughly separated into three groups as shown in Figure 1 and Figure 2: the inner belt at semimajor axis $a < 40$ AU, the outer (classical) belt at $a > 42$ AU, and the “scattered disk.” The boundary between the inner belt and the outer belt is the ν_8 resonance, which is located at 41 AU. The scattered disk is composed of the objects that have been largely scattered by Neptune so that they have large eccentricities and perihelion distance q around 30–35 AU (Duncan & Levison 1997). In the inner belt, semimajor axes of the objects are concentrated near the 3:2 resonance (39.5 AU) with Neptunian orbit, and both eccentricities e and inclinations i are largely excited. Root mean squares ($\langle e^2 \rangle^{1/2}$ and $\langle i^2 \rangle^{1/2}$) of the eccentricities and inclinations are about 0.21 and 0.23, respectively. Objects with low eccentricity ($e < 0.1$) are lacking in this region. In the outer region, semimajor axes of the objects are broadly distributed. The eccentricities of objects in the outer belt are distributed up to 0.2, and their $\langle e^2 \rangle^{1/2}$ is about 0.1. On the other hand, the inclinations are nearly equal to those in the inner belt. They are distributed up to 0.6 and $\langle i^2 \rangle^{1/2} \simeq 0.2$. Highly eccentric ($e \gtrsim 0.2$) orbits in this region may have been dynamically removed, because such orbits have $q \lesssim 35$ AU and are not protected by mean motion resonances (Duncan, Levison, & Budd 1995). We may not be able to

say i is more highly excited than e ; nevertheless, the high i distribution up to ~ 0.6 is notable.

The orbits near secular resonances ν_5 (~ 34 AU), ν_7 (~ 35 AU), and ν_8 (~ 41 AU) are unstable. The eccentricities of objects are excited by these secular resonances, so that q of the orbits becomes smaller than 35 AU to be destabilized (Duncan et al. 1995).

Although objects with low eccentricities and semimajor axes between 36 and 39 AU are stable (Duncan et al. 1995), few objects have been observed in this region. Sweeping of the 3:2 mean motion resonance caused by planetary migration explains the depletion at $36 \text{ AU} < a < 39 \text{ AU}$ as well as overdensity of objects at the 3:2 resonance with high eccentricities (Malhotra 1993, 1995). To excite e up to 0.3, migration length should be about 7 AU. The ν_{18} secular resonance located near the 3:2 resonance would excite inclinations of the objects there (Malhotra 1998).

In the outer belt, however, there is no low-order mean motion resonance or secular resonance. It is hard to excite eccentricity and inclination up to the observed level by present planetary perturbations alone. Here we focus our attention on the origin of eccentricities and inclinations of the KBOs in the outer belt.

Morbidelli & Valsecchi (1997) proposed that Earth-sized large planetesimals were scattered by Neptune into the Kuiper belt, and these large planetesimals could excite the orbits in the outer belt before they were ejected. Based on this model, Petit, Morbidelli, & Valsecchi (1999) did more detailed study, combining restricted three-body calculation and Monte Carlo simulation. This mechanism could provide large eccentricities with large mass depletion in the inner belt and moderate eccentricities in the outer belt but may require appropriate orbital configuration, masses, and ejection timescale of the large planetesimals. Ida, Larwood, & Burkert (2000) proposed hypothetical stellar encounters

¹ <http://cfa-www.harvard.edu/~graff/lists/TNOs.html>.

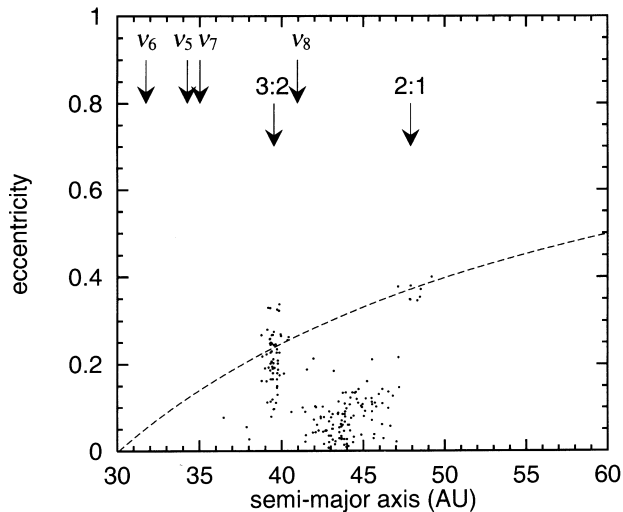


FIG. 1.—Distribution of eccentricities vs. semimajor axes of the present Kuiper belt objects (see URL listed in footnote 1). Locations of important mean motion resonances with Neptune orbit are shown. Locations of the secular resonances, which pump up eccentricity (v_5 , v_6 , v_7 , and v_8), are also shown. Above the dotted line, the orbits of the KBOs cross Neptunian orbit.

as another mechanism to explain $\langle e^2 \rangle^{1/2} \sim 0.1$ in the outer belt. They suggest that the proto-Sun was in a stellar aggregation in the formation stage and experienced stellar encounters with pericenter distance of 100–200 AU. These stellar encounters enhance the eccentricities of KBOs. The acquired eccentricity rapidly increases with semimajor axis. A stellar encounter with pericenter distance ~ 100 –200 AU excites $\langle e^2 \rangle^{1/2}$ to ~ 0.1 in the outer belt while it keeps $\langle e^2 \rangle^{1/2} \lesssim 0.05$ well inside 39 AU. If this encounter is followed by the planet migration as proposed by Malhotra (1993, 1995), this mechanism nicely explains the observed eccentricities both in inner and outer belts. In order for the 3:2 resonance to trap the KBOs efficiently, initial eccentricities of KBOs before trapping must be smaller than 0.05

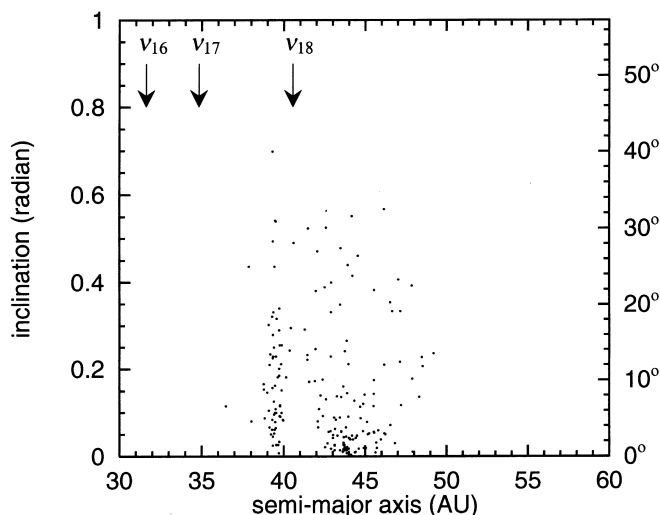


FIG. 2.—Distribution of inclinations vs. semimajor axes of the present KBOs (see URL listed in footnote 1). Inclination is given in radians. Locations of the secular resonances, which pump up inclination (v_{16} , v_{17} , and v_{18}), are also shown.

(Malhotra 1995). If e is pumped up to $\gtrsim 0.05$ at $a \lesssim 39$ AU, the 3:2 resonance fails to trap many objects, and they should remain at $a \lesssim 39$ AU. On the other hand, Hahn & Malhotra (1999) suggest that the 2:1 mean motion resonance, which also migrates during the Neptunian migration, may partially trap the KBOs and excite their eccentricities between the 3:2 and the 2:1 resonances to ~ 0.1 .

These models can explain $\langle e^2 \rangle^{1/2} \sim 0.1$ in the outer belt. As for inclination in the outer belt, however, it is not easy for these models to explain the observed value. In the Hahn & Malhotra model, inclination is not excited. Perturbations of the hypothetical Earth-sized planetesimals may not excite inclinations so largely. The passing stellar encounters can excite inclinations more than the other models; however, optimal choice of parameters of the encounters may be required so that more detailed examination would be needed.

Here we examine another possibility to excite eccentricities and inclinations of KBOs: sweeping secular resonances caused by depletion of the primitive solar nebula. The importance of sweeping secular resonances for eccentricities of planets was first pointed out by Ward, Colombo, & Franklin (1976). Secular resonances occur when the proper frequency associated with the longitude of perihelion or that of the ascending node of an object coincides with one of the eigenfrequencies of planetary precessions. Coincidence of the longitude of perihelion causes excitation of eccentricity, and that of ascending node causes excitation of inclination. When the solar nebula existed, the nebula potential would have significantly altered locations of the secular resonances because even the minimum mass nebula model has more than 10 Jovian mass. As the nebula was depleted, the secular resonances moved from their initial locations to the present ones, pumping up eccentricities and inclinations of field objects. As for the asteroid belt, the sweeping secular resonances successfully explain their large eccentricities (Heppenheimer 1980; Lecar & Franklin 1997) and inclinations (Nagasawa, Tanaka, & Ida 2000).

As for the Kuiper belt, similar sweeping secular resonances are caused by depletion of an initially massive planetesimal disk (Malhotra, Duncan, & Levison 2000). Although Malhotra et al. do not present detailed results, they suggest that the inclinations are excited (~ 0.25 radians) at $a > 42$ AU.

We will hereafter consider sweeping secular resonances caused by depletion of the primitive solar nebula, but the results below can also be applied to a (non-self-gravitating) planetesimal disk if initial surface density is reduced by a factor $\sim 1/100$. The orbits of KBOs are affected mainly by the perturbations of the largest planet Jupiter, the nearest planet Neptune, and the nebula potential. Although most of the nebula may have been depleted when Neptune was formed, only a bit of residual nebula can excite inclinations of the KBOs as well as their eccentricities as shown later. The sweeping secular resonances explain the distribution of inclinations in the outer belt, which is not easily explained by other models.

This paper investigates the effects of sweeping secular resonances in the Kuiper belt through analytical estimation and direct orbital integration, using a three-dimensional model of the primitive solar nebula. In § 2, we describe our models and briefly explain the analytical estimation of the effects of sweeping secular resonances. In § 3, we show the

results of the analytical estimation. The regions swept by secular resonances and the magnitude of excitations of eccentricities and inclinations are analytically derived. In one model, we compare the analytical estimation with direct orbital integration and show a good agreement. Inclinations of the KBOs are largely excited by the sweeping secular resonances during the depletion of the solar nebula, if the residual nebula with 0.1% of the initial surface density is depleted on a timescale of 10^7 – 10^8 yr when present-sized Neptune exists, or 10^8 yr when Neptune does not exist (for eccentricity excitation, Neptune with more than one-fifth of the present mass is needed). In § 4, we give conclusions and discuss the theoretical models and future observations. At present, the other models (large planetesimals, stellar encounter, 2:1 resonance sweeping) for excitation of e and i in the outer belt cannot be ruled out. However, we point out that predicted e and i beyond 50 AU are quite different among the models. Future observations for the region beyond 50 AU will evaluate the models.

2. MODEL AND METHODS

We consider a system that consists of the Sun, four Jovian planets, and massless particles (which correspond to KBOs) in the nebula and study the effects of secular resonances on orbital evolution of the test particles according to depletion of the nebula. We assume the nebula is cylindrically symmetric around the Sun and also symmetric with respect to the ecliptic plane. We consider the nebula as a gravitational potential source and neglect hydrodynamic gas drag to objects because the damping timescale of eccentricities and inclinations due to gas drag are longer than 10^8 yr in the minimum mass solar nebula (see § 2.2.1) for objects with mass larger than 10^{21} g and $e = i = 0.1$. For a massive nebula, this timescale can be shorter. As shown later, however, the nebula density is lower than that in the minimum mass solar nebula model when secular resonances pass through the Kuiper belt to excite the eccentricities and inclinations of the KBOs. The damping timescale is thus much longer than the nebula lifetime ($\sim 10^6$ – 10^7 yr) inferred by observations (Strom, Edwards, & Skrutskie 1993; Zuckerman, Forveille, & Kastner 1995), so that the gas drag hardly affects orbits of the Kuiper belt objects. The nebula affects the orbits through precessing the longitudes of perihelia and ascending nodes by its gravitational potential.

2.1. Sweeping Secular Resonances

We briefly explain the sweeping secular resonances (for details, see, e.g., Ward et al. 1976; Heppenheimer 1980; Ward 1981). Secular resonances occur when the proper frequency associated with the longitude of perihelion or that of the ascending node of a particle coincides with one of the eigenfrequencies of planetary precessions. The former causes excitation of eccentricity, and the latter excites inclination. Considering the Sun and four Jovian planets in the present solar system configuration, seven secular resonances exist beyond the orbit of Neptune. The resonances of four eccentricities are at 31.7, 34.3, 35.0, and 41.0 AU from the Sun. They are called the ν_6 , the ν_5 , the ν_7 , and the ν_8 resonances, respectively. Resonances of inclinations exist at 31.6, 34.8, and 40.6 AU, which are called the ν_{16} , the ν_{17} , and the ν_{18} resonances, respectively. If a nebula exists, its gravitational potential changes the locations of the secular

resonances. As the nebula is depleted, the secular resonances migrate. This was first pointed out by Ward et al. (1976). They estimated the excitation of e and i of Mercury with the linear theory. Assuming that e and i of objects are small, they obtained the magnitude of excitation of e and i from zero by a single passage of secular resonance j as

$$\Delta e = \sqrt{2\pi\eta_j} \left[\frac{d(g - g_j)}{dt} \right]^{-1/2}, \quad (1)$$

$$\Delta i = \tan^{-1} \left\{ \sqrt{2\pi\mu_j} \left[\frac{d(f - f_j)}{dt} \right]^{-1/2} \right\}, \quad (2)$$

where g_j and f_j are eigenfrequencies of the planetary precessions, η_j and μ_j are amplitudes of forced oscillations, and g and f are the change rates of perihelion and ascending node of a particle. The locations of secular resonances ν_j are given by $g = g_j$ or $f = f_j$. We assign eigenfrequencies g_j ($j = 5, 6, 7, 8$) as $g_8 < g_7 < g_5 < g_6$ and call the corresponding resonances ν_8 , ν_7 , ν_5 , and ν_6 . They are the resonances of eccentricity. We assign f_j as $f_{16} < f_{17} < f_{18} < f_{15} \leq 0$ and call the corresponding resonances ν_{16} , ν_{17} , ν_{18} , and ν_{15} . They are the resonances of inclination. In many cases, an eigenfrequency is close to the change rate of perihelion or ascending node of a particular planet. For example, g_5 is close to the change rate of Jovian perihelion in the present planetary configuration without a nebula. In this case, we say “the ν_5 is the Jovian resonance” or “Jupiter causes the ν_5 resonance.” Because we define the labels of resonances as above, a resonance does not always correspond to the same planet. When the nebula potential changes with time, the correspondence between individual resonances and planets can change. For example, before nebula depletion g_5 is close to the change rate of perihelion of Neptune rather than that of Jupiter; that is, the ν_5 is Neptunian resonance. According to potential change due to nebula depletion, the ν_5 changes from Neptunian resonance to Jovian resonance.

We investigate the sweeping secular resonances in KBOs with both direct orbital integration and an analytical method, following Nagasawa et al. (2000), who studied the sweeping secular resonances in the asteroid belt. In the analytical method, we determine the locations of secular resonances at every stage of the nebula depletion. The disturbing functions are obtained by numerical integration of expanded three-dimensional nebula potential that will be modeled in § 2.2.1. With given depletion speed of the nebula, we estimate the excitations of e and i using equation (1) and equation (2) at the locations where the resonances take place. Summing up all the effects of resonances’ passages, we obtain the total (maximum) excitations of e and i at each semimajor axis.

Nagasawa et al. (2000) showed that direct orbital integrations gave consistent results with the analytical estimation. In the case of the present-day model for the KBOs (see § 2.2), we did the numerical orbital simulations and confirmed that the results are also consistent in the KBO case. In the direct orbital integrations, we use the time-symmetrized Encke’s method (Nagasawa et al. 2000), which calculates evolution of apse and node accurately. We put four Jovian planets with velocities and positions at JED 2,449,899.5 and present masses and put 21 massless particles between 40 and 60 AU as KBOs. According to Nagasawa et al. (2000), the three-dimensional nebula potential is directly calculated at lattice points in the r - z plane. We

integrate orbits, interpolating the potential by bicubic interpolation.

We mainly use the analytical method here, with general features that are easier to understand.

2.2. Modeling

2.2.1. Nebula Model

We use several different nebula models. For a standard nebula model, we adopt the minimum mass solar nebula model (Hayashi 1981), in which the distribution of gas density is given by

$$\rho_H = 1.4 \times 10^{-9} \left(\frac{r}{1 \text{ AU}} \right)^{-k} \times \exp \left[- \left(\frac{z}{0.047(r/1 \text{ AU})^{5/4}} \right)^2 \right] \text{ g cm}^{-3}, \quad (3)$$

where $k = -11/4$ and (r, z) are cylindrical coordinates centered at the Sun. We assume that nebula temperature is isothermal in the z -direction. We extrapolate equation (3) to 150 AU, where the distribution of the nebula gas is truncated. If the location of the truncation is farther from the Kuiper belt region, the location of the truncation does not matter. In most cases, we assume a nebula gap along the orbit of Jupiter with half-width of 1.233 AU owing to tidal interaction between nebula and Jupiter (e.g., Lin & Papaloizou 1985, 1993). In some cases, we also consider the cases without a nebula gap. We also study other powers of the nebula density ($k = 15/4, 13/4, 9/4$, and $7/4$). These correspond to radial gradient of surface density Σ , which is obtained by vertical integration of equation (3), as $\Sigma \propto r^{-k+5/4} = r^{-k'}$, $k' = 5/2, 2, 1$, and $1/2$. In shallow power cases (k or k' is small), the initial surface density in the Kuiper belt is larger than that the minimum mass nebula model. Because secular resonances sweep the Kuiper belt when the surface density is reduced to a value that is much smaller than that in the minimum mass nebula model, the magnitude of the initial surface density does not affect the results. However, relative magnitude at different positions, which is determined by k or k' , affects the results.

Nebula density is decreased from given initial distribution. We consider a uniform nebula depletion model that assumes nebula density ρ is decreased with time t as

$$\rho(r, z, t) = \rho(r, z, 0) \exp(-t/\tau), \quad (4)$$

independent of (r, z) , where τ is a constant parameter. Nagasawa et al. (2000) suggested that inclinations of asteroids become large enough to be comparable to their observational level, when the nebula is depleted nonuniformly in the region between 5 and 10 AU, such as the depletion caused by gap formation owing to tidal interaction between Jupiter and the nebula. However, the nebula is hardly depleted beyond 30 AU by this gap formation mechanism. The outer nebula disk should be depleted by other mechanisms, e.g., viscous diffusion or photoevaporation (Shu, Johnstone, & Hollenbach 1993). Photoevaporation would be effective in the outer region. Hollenbach, Yorke, & Johnstone (2000) suggested that the entire minimum mass solar nebula is depleted on a timescale of the order of 10^7 yr by a combination mechanism of photoevaporation and viscous spreading. For simplicity, we here consider a simple depletion model, i.e., uniform depletion model.

Assuming equation (4), equation (1) and equation (2) can be written as

$$\Delta e = \sqrt{2\pi\eta_j} \left[\frac{d(g - g_j)}{du} u \right]^{-1/2} \sqrt{\tau}, \quad (5)$$

$$\Delta i = \tan^{-1} \left\{ \sqrt{2\pi\mu_j} \left[\frac{d(f - f_j)}{du} u \right]^{-1/2} \sqrt{\tau} \right\}, \quad (6)$$

where u is the nebula depletion factor defined as $u \equiv \rho(r, z, t)/\rho(r, z, 0) = \exp(-t/\tau)$.

2.2.2. Model for Planetary System

We consider three models of a configuration of the planets. Since the positions and the masses of planets affect the locations of secular resonances, it is necessary to consider different planetary configurations. The first model assumes present-day masses and orbits of Jupiter, Saturn, Uranus, and Neptune. The semimajor axes are 5.20, 9.55, 19.2, and 30.1 AU, respectively. The eccentricities are 0.0482, 0.0539, 0.0513, and 0.00496. The inclinations are 0.0228, 0.0434, 0.0135, and 0.0309 in radians. We call this model the “present-day model.” We will show that in this model the inclination’s resonances are more effective than the eccentricity’s resonances in the Kuiper belt.

The nebula may have been significantly depleted before completion of accretion of Uranus and Neptune because their core accretion is relatively slow and they do not have massive gas envelopes while their core masses are large enough to initiate runaway gas accretion (Mizuno 1980; Bodenheimer & Pollack 1986). Thereby, the second model assumes proto-Uranus and proto-Neptune with masses smaller than the present ones. We call this model the “small-mass model.” We consider two cases: the masses of Uranus and Neptune are one-fifth and 1/100 of the present masses. The masses of Jupiter and Saturn and the semimajor axes of four planets are taken as the present ones. As shown later, a longer timescale of the nebula depletion is required to obtain large eccentricity and inclinations with this model.

Fernández & Ip (1984) and Hahn & Malhotra (1999) suggested separations between the planetary orbits gradually expand as the planets scatter the planetesimals. This means that the semimajor axes of the planets may have been different from present-day ones. In the third model, the “compact model,” the semimajor axis of four Jovian planets are 5.4, 8.7, 16.3, and 23.2 AU, according to the results of Fernández & Ip (1984). For planetary masses, the present values are taken. We do not change the semimajor axis of planets during the estimation for simplicity (see discussion in § 3.3). The secular resonances basically occur at smaller semimajor axes than in the case of the present-day model.

3. RESULTS

3.1. The Case of the Present-Day Model

The semimajor axes of four Jovian planets are set to be 5.20, 9.55, 19.2, and 30.1 AU and their masses are 318, 95.2, 14.5, and 17.2 $M_\oplus$, respectively. The starting nebula is the minimum mass solar nebula described by equation (3).

We first show the results of linear analysis. Figure 3 shows the values of eigenfrequencies g and g_j , where g is the rotation rate of longitude of perihelion ($\dot{\omega}$) of a particle caused by perturbations of the nebula and planets and g_j 's are eigenfrequencies of the planets. g_6, g_5, g_7 , and g_8 are

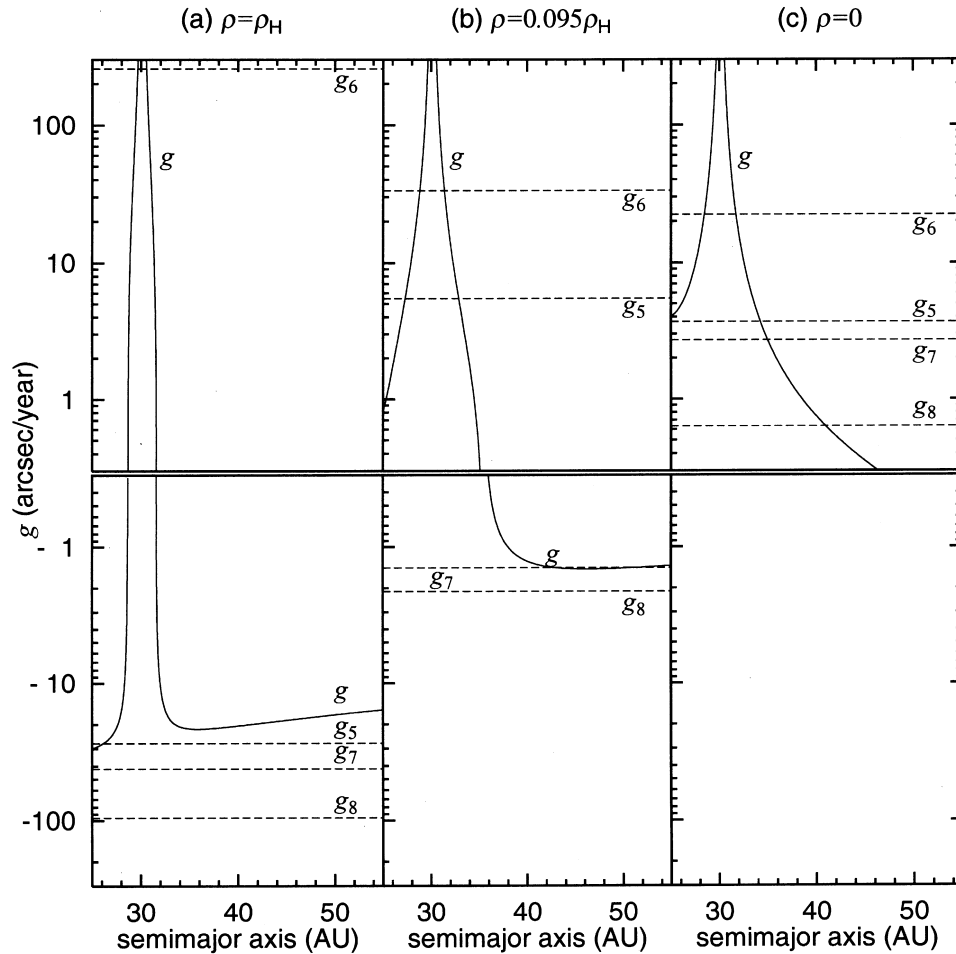


FIG. 3.—Precession frequency g of longitude of perihelion of a massless particle as a function of semimajor axis a . Eigenfrequencies g_j of that of the planetary system are also shown. (a) The case of initial nebula (the minimum mass solar nebula is assumed). g and g_j do not coincide with each other in the region of the Kuiper belt. (b) The case in which the nebula density decreases to 9.5% of the initial density (ρ_H). Secular resonances in the Kuiper belt occur at the positions where g crosses g_7 (42.8 and 51.4 AU). (c) The case in which there is no nebula (in other words, the present state). Secular resonances occur at 31.7, 34.3, 35.0, and 41.0 AU.

257, -27, -42, and -95 arcsec yr⁻¹, and they nearly correspond to the rotation rates of perihelia of Jupiter, Neptune, Uranus, and Saturn, respectively. Perturbations of the nebula become strong as decrease in heliocentric distance, since the nebula density increases toward the Sun. Perturbations of a planet become strong as a object approaches the planet. Hence, g depends on the semimajor axis. Perturbations of the nebula around the particle make $\tilde{\omega}$ regress, while those of the planets (and also distant part of the nebula) make it progress. Figures 3a, 3b, and 3c show the cases of nebula density $\rho = \rho_H$, $\rho = 0.095\rho_H$, and $\rho = 0$. As the nebula density decreases, the perturbations of the nearby nebula diminish, so that g and g_j 's move upward. With the starting nebula density (ρ_H), the rotation rate of $\tilde{\omega}$ of a particle is negative except in the vicinity of the planets. The rotation rates (g_j 's) of planetary $\tilde{\omega}$ are all negative except for g_6 corresponding to Jupiter. Since we are considering the nebula gap along Jovian orbit, Jovian perihelion does not regress but progress. If there is no nebula gap, all g_j 's are negative, i.e., all planetary $\tilde{\omega}$ regress. In this case, g_6 , g_5 , g_7 , and g_8 are -27, -41, -68, and -154 arcsec yr⁻¹, and they nearly correspond to the rotation rates of perihelia of Neptune, Uranus, Saturn, and Jupiter, respectively.

All of the absolute values of g_j 's are larger than that of g in the Kuiper belt, so that the secular resonances do not occur there until the nebula density decreases. When the nebula density sufficiently decreases, g_j 's move upward more than g in the Kuiper belt because the nebula density is higher and has larger effect in planetary regions than in the Kuiper belt. Then g_j 's catch up with g , and the secular resonances occur. When $\rho \simeq 0.18\rho_H$, g_5 starts crossing g (the ν_5 secular resonance). The secular resonances occur at about 42 AU at first and start moving both directions of outward and inward. g also contacts with g_7 (the ν_7 secular resonance) at 46 AU when $\rho = 0.097\rho_H$. In the case of $\rho = 0.095\rho_H$ (Fig. 3b), g crosses g_7 at 42.8 and 51.4 AU. Thus, the eccentricity's resonances sweep the main region of the Kuiper belt from outside to inside. Figure 4 shows the locations of eccentricity's secular resonances as a function of the nebula depletion factor u [$u = \rho/\rho_H = \exp(-t/\tau)$]. When $\rho = 0.18\rho_H$, $0.097\rho_H$, and $0.040\rho_H$, the ν_5 , ν_7 , and ν_8 secular resonances begin to sweep the Kuiper belt region, respectively. The mass of nebula between 30 and 50 AU is about $230 M_\oplus$ at $\rho \sim 0.18\rho_H$. The eccentricity's resonances sweep the Kuiper belt from outside to inside and nearly settled to the present locations when the nebula density decreases to 1/100 of the initial value. If there is not a nebula gap, all of

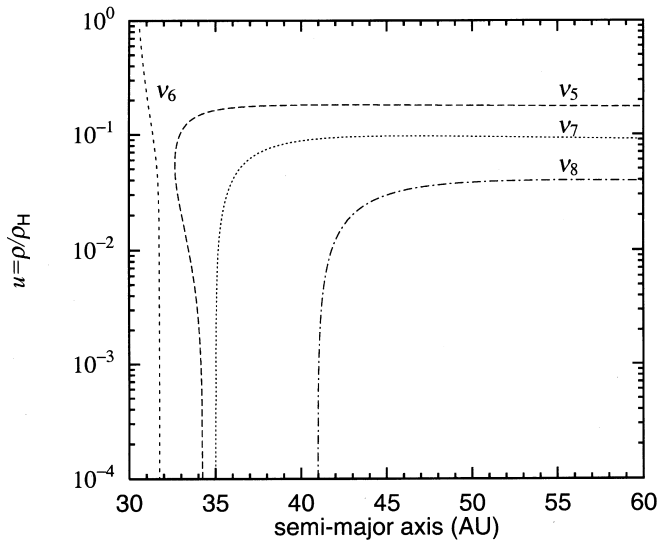


FIG. 4.—Locations of the secular resonances v_5 , v_6 , v_7 , and the v_8 in the present-day model, as a function of nebula depletion factor u . The depletion factor u is defined by $u = \rho/\rho_H = \exp(-t/\tau)$, where ρ is the nebula density on the way of depletion, ρ_H is the initial nebula density of the minimum mass solar nebula, t is time from beginning of the nebula depletion, and τ is a characteristic depletion timescale. In this figure, we assume a fixed nebula gap with half-width 1.233 AU along Jovian orbit throughout the nebula depletion. The v_5 , v_6 , v_7 , v_8 resonances are almost caused by Neptune, Jupiter, Uranus, and Saturn, respectively, in the initial state. In the depleted nebula case, they are almost caused by Jupiter, Saturn, Uranus, and Neptune, respectively.

the eccentricity's resonances including v_6 sweep the Kuiper belt.

Figure 5 shows the change rate f of ascending node of a particle as a function of semimajor axis. Eigenfrequencies (f_j) of the change rate of ascending nodes of planets are also plotted. Figures 5a and 5b show the cases of $\rho = \rho_H$ and $\rho = 0$, respectively. Both the nebula's and the planets' gravitational forces regressively rotate the ascending node Ω of the particle and the planets. Ω regresses more rapidly as the

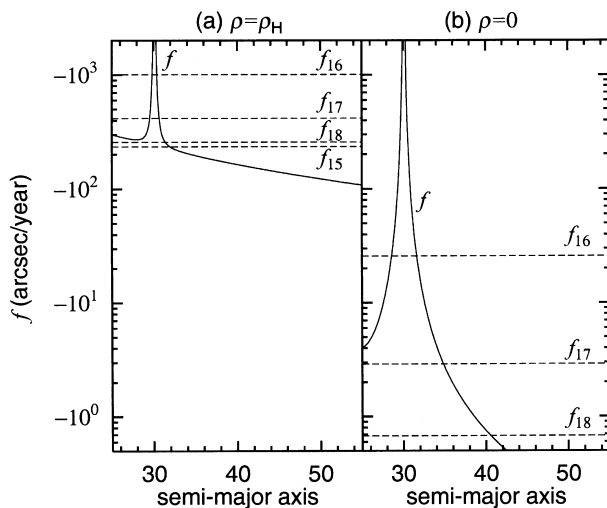


FIG. 5.—Precession frequency f of longitude of ascending node of a massless particle is shown with eigenfrequencies f_j of that of the planets. (a) The initial state where the nebula density is ρ_H . f and f_j coincide with each other only in the vicinity of Neptune. (b) The final state with zero nebula density (in other words, the present state). Secular resonances occur at 31.6, 34.8, and 40.6 AU.

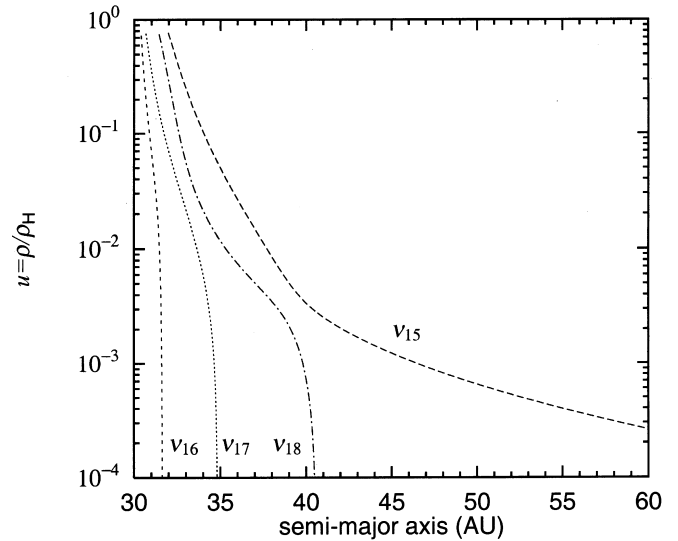


FIG. 6.—Locations of the secular resonances v_{15} , v_{16} , v_{17} , and v_{18} in the present-day model, as a function of nebula depletion factor u . The v_{15} , v_{16} , v_{17} , v_{18} resonances are almost caused by Neptune, Jupiter, Uranus, and Saturn, respectively, in the initial state. In the depleted nebula case, they are almost caused by Jupiter, Neptune, Uranus, and Saturn, respectively.

semimajor axis of the particle decreases. In the Kuiper belt, the nebula regresses Ω of planets faster than that of the particle. In the vicinity of Neptune, however, Neptunian perturbations regress the particle's Ω faster than the planets' Ω . Inclination resonances occur only in the vicinity of Neptune. As the nebula density decreases, the difference between f caused by the nebula and f_j 's decreases, and f_j 's and f meet at points not so close to Neptune. As a result, the inclination's resonances migrate from the vicinity of Neptune to the present location, i.e., from inside to outside. Figure 6 shows the locations of inclination's secular resonances as a function of the degree of nebula depletion. The v_{15} and v_{18} resonances, which correspond to f_5 and f_8 , respectively, sweep the Kuiper belt from inside to outside. The v_{15} starts to pass the Kuiper belt when the nebula density decreases to $1/250$ of the initial density and goes to infinity across the Kuiper belt when the nebula density becomes $1/5000$ of the initial density. Therefore, both the eccentricity and inclination resonances sweep the Kuiper belt after nebula density decreases below 10^{-2} of the initial value. As shown below, such a small amount of the residual nebula can be responsible for high eccentricities and inclinations of the KBOs.

In Figure 7 and Figure 8, the excited magnitude of eccentricity and inclination predicted by equations (5) and (6) are plotted. A nebula depletion timescale $\tau = 10^7$ yr is assumed here. For other τ , Δe and Δi are multiplied by $(\tau/10^7 \text{ yr})^{1/2}$ according to equation (5) and equation (6). The values of η_j and μ_j are determined by orbits of current Jovian planets. If eccentricities and inclinations of all Jovian planets are half of the present ones, the resultant Δe and Δi are also half of the above estimate. Note that the sweeping secular resonances affect e and i in either a constructive or destructive way when the free eccentricity or inclination are not initially zero. The solid thick lines show the total excitation in eccentricity and inclination as a summation of excitations by sweeping of individual resonances, assuming that all of them work constructively. These lines correspond to the

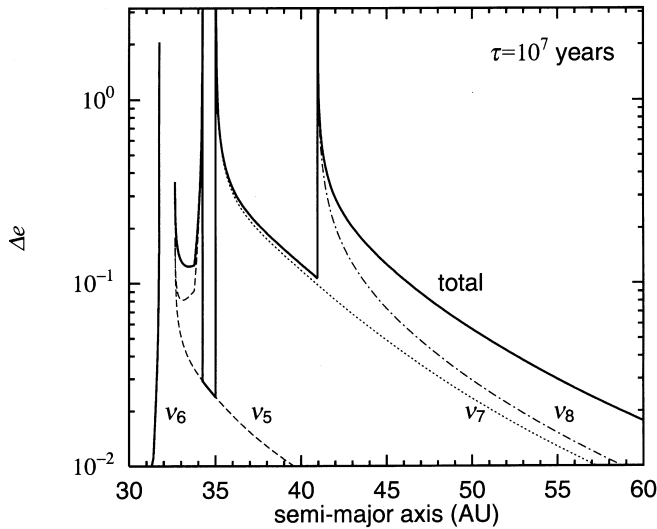


FIG. 7.—Estimated excitations in eccentricity in the present-day model for depletion timescale $\tau = 10^7$ yr. Dashed lines show the excitation due to single sweepings of the v_5 , v_6 , v_7 , and v_8 resonances. The solid thick line shows the total excitation in eccentricity as a summation of individual excitations. This line corresponds to the maximum excitation in the case that all the resonances work constructively. The result for other τ is obtained by multiplying the vertical scale by $(\tau/10^7 \text{ yr})^{1/2}$.

maximum excitations. Nevertheless, they give a good estimate, since a certain resonance usually dominates the others as shown in Figure 11 and Figure 12. The estimate is consistent with the numerical simulations as will be shown below.

Figure 7 and Figure 8 show that inclination is pumped up more highly than eccentricity for the same nebula depletion timescale, since the inclination's resonances migrate slower than the eccentricity's resonances. The v_{18} resonance is the most important for the inclination excitation near the 3:2 resonance ($a \approx 39.5$ AU), while the v_{15} resonance is the most important in the outer belt ($a \gtrsim 42$ AU). The v_{15} reso-

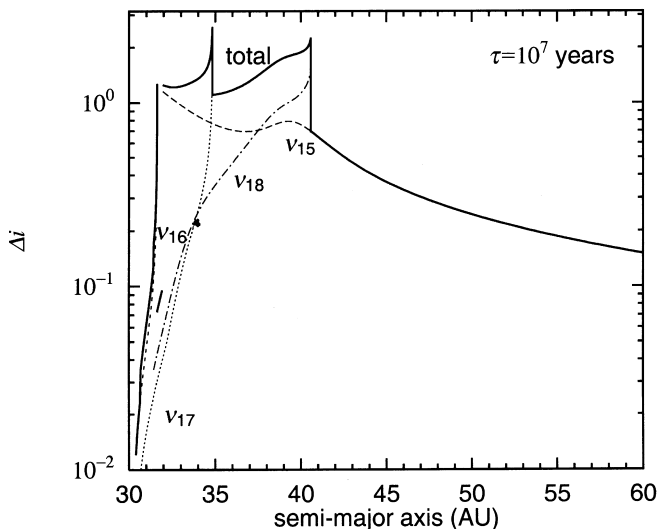


FIG. 8.—Estimated excitations in inclination in the present-day model for depletion timescale $\tau = 10^7$ yr. Dashed lines show the excitation due to single sweeps of the v_{15} , v_{16} , v_{17} , and v_{18} resonances. The solid thick line shows the maximum excitation in inclination as a summation of individual excitations. The result for other τ is obtained by multiplying the vertical scale by $(\tau/10^7 \text{ yr})^{1/2}$.

nance is closely related to the regression speed (f_5) of the Jovian ascending node. Jupiter is essentially responsible for excitation of inclination in the outer belt.

In the early stage of the nebula depletion, Jupiter, Saturn, and Neptune cause the v_6 , v_8 , and v_5 , respectively. They cause the v_5 , v_6 , and v_8 , respectively, in the depleted nebula. Uranus causes the v_7 throughout the depletion. We defined the labels of the resonances as in § 2.1, so that the driving planets for the individual resonances can change between before and after nebula depletion. Although the perihelion of a particle regresses in the Kuiper belt, Jovian perihelion progresses since we are considering the nebula gap along the Jovian orbit, so that the v_6 resonance does not sweep the region beyond 34 AU (Fig. 3 and Fig. 4). Then, the resonance caused by the second largest planet, Saturn, is the most important for excitation of eccentricity in the outer belt. Saturnian resonance is the v_8 when it passes through the Kuiper belt. In the late stage of the nebula depletion, the v_8 becomes Neptunian resonance and settles into 41 AU. It does not sweep the inner belt. On the other hand, the v_7 resonance is important for excitation of eccentricity at $a \lesssim 40$ AU. As a result, in the late stage of nebula depletion, the v_7 and v_8 resonances are important, since they are located near Kuiper belt.

If there is no nebula gap, the Jovian resonance also passes through the Kuiper belt. The Jovian resonance is 1.5 times as strong as the Saturnian resonance. The total increment of eccentricity is about twice as large as that in the case without the nebula gap. On the other hand, since the nebula gap does not change the rotational direction of Jovian ascending node, the overall features of sweeping of the inclination's resonances do not depend on nebula gap. As a result, the increment of the inclination hardly changes.

Although we are considering uniform depletion, the results in the cases of nonuniform depletion are predicted to some degree with Figure 3 and Figure 5. For example, consider the case in which the nebula is depleted earlier in $a < 35$ AU than in $a > 35$ AU. This depletion is analogous to that of a planetesimal disk. In the region of $a \lesssim 35$ AU, large planets exist, and their gravity ejects planetesimals, while such ejection does not occur at $a > 35$ AU. The nebula outside 35 AU is mainly responsible for the precession of perihelion and ascending node of a particle in the Kuiper belt (g and f , respectively). On the other hand, the nebula inside 35 AU is mainly responsible for the eigenfrequencies of the planetary precessions (g_j and f_j). In the stage of inner nebula depletion, g does not change so largely from that in Figure 3a while g_j 's change from those in Figure 3a to those in Figure 3c. The secular resonances v_5 , v_7 , and v_8 migrate from 36 AU to both sides. The one settles to the vicinity of Neptune, and the other sweeps the Kuiper belt outward. If there is no nebula gap, the v_6 migrates in the same way as the other resonances. During the next stage of outer nebula depletion, g_j 's stay there but g changes from that in Figure 3a to that in Figure 3c. Then the secular resonances migrate outward from the vicinity of Neptune to present locations. Note that in the case of a planetesimal disk, ρ starts from $\sim 0.01\rho_H$. Then, the early sweeping throughout the Kuiper belt does not exist, and eccentricities are not pumped up in the outer belt. Similarly, the evolution of inclinations' resonances is predicted with Figure 5. According to the depletion of the inner nebula, the resonances v_{15} , v_{17} , and v_{18} once sweep the Kuiper belt from inside to outside. During the depletion of the outside

nebula, they return to the present locations except for the v_{15} (it does not return to the Kuiper belt since $f_5 \sim 0$). The v_{16} resonance does not occur beyond 35 AU because f near 35 AU also changes somewhat in the stage of the depletion of the inner nebula. Although the way and time of passage of the secular resonances change, the total excitation in eccentricities and inclinations do not change so largely. In a similar way, we can also predict the resonances' migration in the reverse case that the nebula beyond 35 AU is depleted earlier, which might correspond to photoevaporation.

Next we consider the cases of other nebula models. We will show that the results do not change except for the case in which the radial gradient of surface density is too gentle. The values of Δe and Δi are shown in Figures 9 and 10 for different initial distributions of nebula density: k values of equation (3) are taken as 15/4, 13/4, 11/4, 9/4, and 7/4. These correspond to surface density gradient with the power of $k' = 5/2, 2, 3/2, 1$, and $1/2$, respectively, if we assume equation (3). In the standard case, $k = 11/4$ and $k' = 3/2$. Note that the paths of resonance sweeping are changed by only relative nebula density between different positions (i.e., k or k') but not by the magnitude of initial nebula density. The increments of eccentricity and inclination are almost the same for $k' = 5/2, 2, 3/2$, and 1 . For $k' = 1$, the v_{15} resonance sweeps only the region beyond 40 AU, and the total excitation of i is slightly smaller than that in the case of $k' = 5/2, 2$, and $3/2$. On the other hand, in the case of $k' = 1/2$, the v_{15} resonance, which is the most important for excitation of i in the other cases, does not appear in the Kuiper belt region. This is because the difference in nebula perturbation between the Kuiper belt and region of Jupiter is too small to coincide with the rotation speeds of the ascending nodes of particles there. The sweeping direction of eccentricity resonances in the cases of $k' = 1/2$ differs from that in the cases of $k' \geq 1$. In the case with $k' = 1/2$, eccentricity and inclination are excited only in the inner belt region. Except for $k' = 1/2$, however, the features of resonance sweeping and resultant Δe and Δi are very similar.

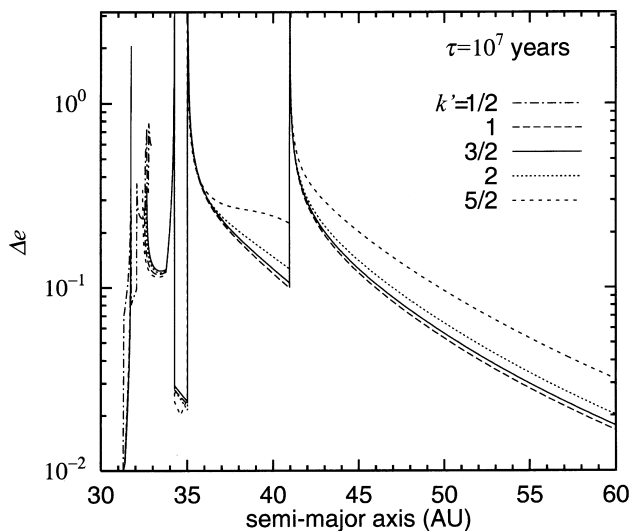


FIG. 9.—Estimated total excitations in eccentricity for various radial gradients of nebula surface density. The present-day model and the nebula depletion time scale of $\tau = 10^7$ yr are adopted here. Solid line shows the case of density distribution ($\Sigma \propto r^{-k}$, $k' = 3/2$) (in the minimum mass solar nebula model). Dash-dotted, long-dashed, dotted, and short-dashed lines represent the case of $k' = 1/2, 1, 2, 5/2$, respectively. Except for the case of $k' = 1/2$, the excitation magnitude is similar in each case.

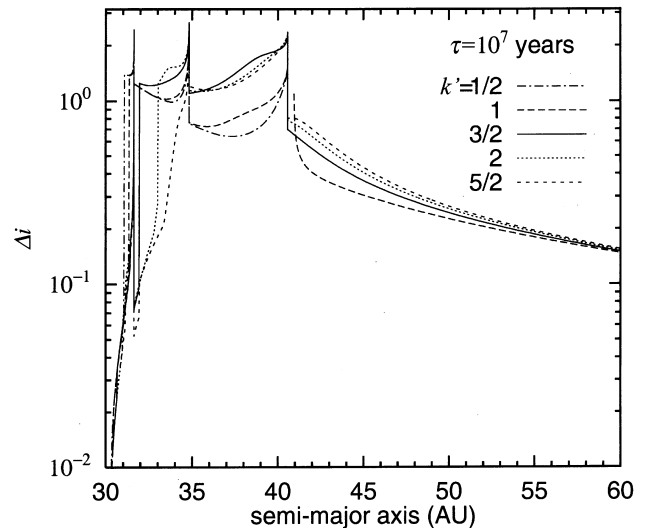


FIG. 10.—Estimated excitations in inclination for various radial gradients of nebula surface density. The present-day model and the nebula depletion timescale of $\tau = 10^7$ yr are adopted here. Meanings of the lines are the same as those in Fig. 9. Except for the case of $k' = 1/2$, the excitation magnitude is similar in each case.

The results do not depend largely on the nebula surface density gradient k' for $k' \geq 1$, including $k' = 3/2$ close to that in the minimum mass nebula model.

We also performed direct orbital integration and compared the results with the above linear analysis, in the case of the minimum mass solar nebula model ($k' = 3/2$). The orbits of test particles are integrated, directly calculating the three-dimensional nebula potential according to the methods of Nagasawa et al. (2000). We initially put 21 massless particles between 40 and 60 AU with equal intervals of 1 AU with $e = i = 0.001$. Other orbital elements are taken randomly. Orbital elements at JED 2,449,899.5 are adopted as initial conditions of Jovian planets. The nebula depletion timescales are $\tau = 10^5, 10^6$, and 10^7 yr. We integrated the orbits for at least the period of 10 times of τ . In Figure 11 and Figure 12, we plot the maximum values of e and i after the secular resonances pass through the region. Analytically estimated Δe and Δi in total are also plotted corresponding to each τ in these figures. Roughly speaking, excitations of eccentricities and inclinations become larger as the nebula depletion timescale becomes longer. The analytical results coincide with the numerical ones except for the effects that are not taken into account in the analytical estimations, as explained below (also see Fig. 13). In the numerical results, eccentricities are excited to be more than 0.05 in the entire region by forced oscillations; they are highly excited by the 2:1 mean motion resonance near $a \approx 47.9$ AU. In the case of $\tau = 10^7$ yr, numerical results deviate from analytical estimations; in particular, inclinations are not excited highly. This is because Neptunian eccentricity and inclination are changed from the present-day values by passages of secular resonances through Neptunian orbit before secular resonances sweep the Kuiper belt region. If we substitute the changed Neptunian eccentricity and inclination into equation (5) and equation (6), we reproduce the numerical results. As the “present-day” model, we should adopt such initial conditions that final orbital elements of the planets agree with present ones. For this purpose, we performed a run that starts from $\rho/\rho_H \sim 0.003$ with JED 2,449,899.5. In

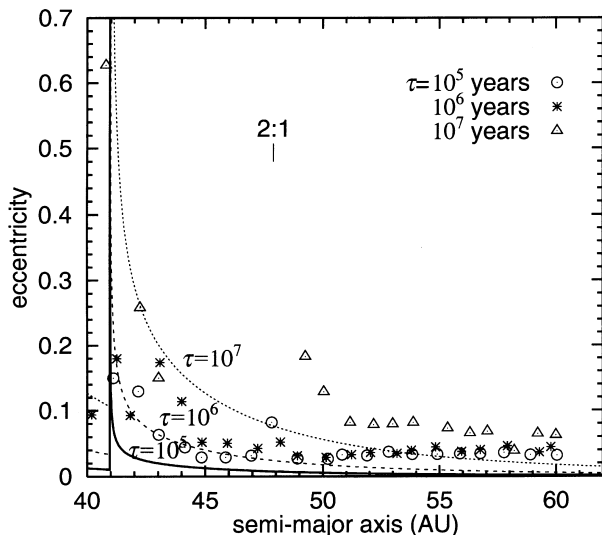


FIG. 11.—Final eccentricities vs. final semimajor axes in the present-day model. The cases with five nebula depletion timescale, $\tau = 10^5$, 10^6 , and 10^7 yr, are plotted. Symbols show the results of direct orbital integrations. Lines show the theoretically estimated total maximum eccentricities. Solid, short-dashed, and dotted lines represent the estimations for $\tau = 10^5$, 10^6 , and 10^7 yr, respectively. The ν_8 resonance now exists at 41 AU.

this case, the resonances already have passed through planetary regions, so that planetary eccentricities and inclinations hardly change. In Figure 13, we plot the final values of time-averaged inclinations after the nebula density becomes sufficiently small. In this case, the inclinations of particles are excited as highly as analytical estimation with the present-day planetary eccentricities and inclinations. In a similar way, if we start a run from $\rho/\rho_H \sim 0.1$ at which the main eccentricity's resonances already passed through planetary regions (but are yet to pass the Kuiper belt region), the eccentricities are not excited so largely as in Figure 11, and their excitation magnitudes coincide with analytical estimations. Thus the analytical estimations on excitations of eccentricities and inclinations of particles agree well with numerical results.

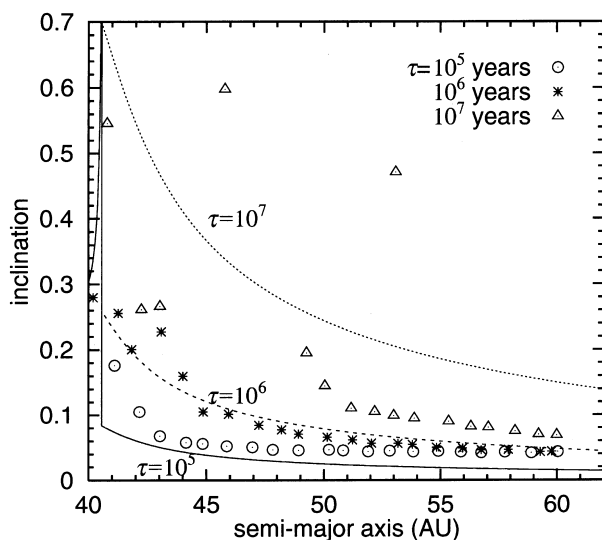


FIG. 12.—Final inclinations vs. final semimajor axes in the present-day model. The cases with five nebula depletion timescales are plotted. Lines are the theoretical maximum inclinations as in Fig. 11.

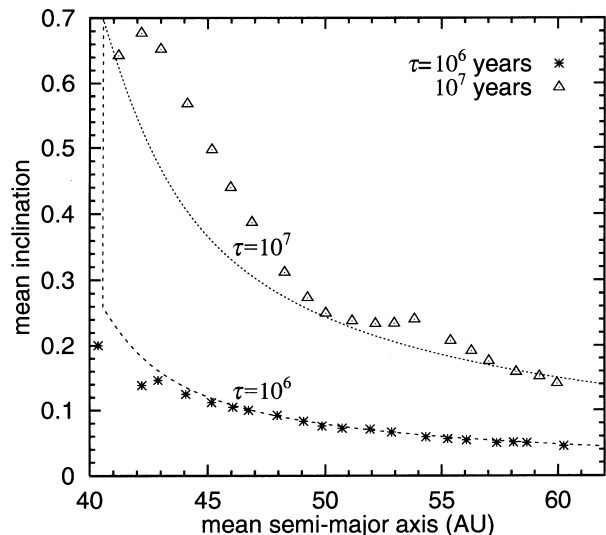


FIG. 13.—Final mean inclinations vs. final mean semimajor axes in the present-day model. Orbital integrations are started from $\rho/\rho_H \sim 0.003$ with present planetary orbital elements. Since the resonances already have passed through planetary regions, the planetary eccentricities and inclinations hardly change. The inclinations of particles agree with the analytical estimation, since the planetary eccentricities and inclinations are not changed by the secular resonances.

3.2. The Case of the Small-Mass Model

Although the present mass of Neptune is smaller than that of Jupiter, Neptunian perturbations to KBOs may be as important as Jovian ones, since Neptune is closer to KBOs than Jupiter. Since the formation timescales of Uranus and Neptune may be longer than 10^7 yr, their growth may be yet to be completed when the sweeping secular resonances occur. It is worth considering the case that Neptune and Uranus are smaller than the present sizes. If proto-Neptune has a smaller mass than the present one, the secular resonances are located closer to Neptune. The perturbations between planets do not vary so largely, since the effects of Uranus and Neptune for g_j and f_j are weak even if they have the present masses. On the other hand, perturbations between Neptune and KBOs become weak when the Neptunian mass is small. Then, for the precession rate of KBO to coincide with that of the planets, the KBOs need larger planetary perturbations, that is, they need to be located closer to Neptune. In Figure 3 and Figure 5, the peak width at Neptune's location becomes small and cross points move to Neptune.

Figure 14 and Figure 15 show the excitation of e and i (Δe and Δi), respectively, in the cases that the masses of Uranus and Neptune are both one-fifth or 1/100 of the present values. These figures show that the secular resonances occur at smaller semimajor axes for smaller masses of planets. The existence of Uranus and Neptune is not so important for excitation of inclination because only the Jovian resonance ν_{15} is important. On the other hand, for the eccentricity excitations, ν_7 and ν_8 , which correspond to Uranian and Neptunian resonances in the late stage of nebula depletion, are the most important as shown in Figure 7. The ν_5 resonance is weak since we consider nebula gap around Jupiter's orbit. Even if there is not a nebula gap, the excitation caused by the ν_5 resonance is the same order of that caused by ν_8 . Accordingly, the excitation of eccentricity depends on the masses of Uranus and Neptune. For

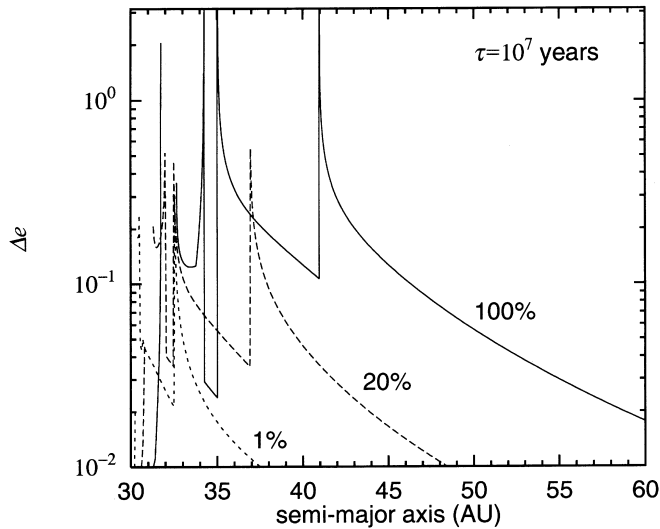


FIG. 14.—Estimated excitations in eccentricity in the small-mass model. The nebula depletion timescale is $\tau = 10^7$ yr. The solid line shows the cases in which Uranus and Neptune have present masses (present-day model). The long-dashed and short-dashed lines show the case in which the Uranian and Neptunian masses are 20% and 1% of the present ones, respectively.

$\langle e^2 \rangle^{1/2} \sim 0.1$ in the outer belt within a reasonable timescale ($\tau \lesssim 10^8$ yr), it is necessary that Uranus and Neptune have at least 20% of the present mass before the nebula is completely depleted.

3.3. The Case of the Compact Model

When the configuration of planets is more compact than the present one, the secular resonances occur at smaller semimajor axes. We show in Figure 16 and Figure 17 the excitation of e and i (Δe and Δi) in the case that the semimajor axes of Jupiter, Saturn, Uranus, and Neptune are 5.4, 8.7, 16.3, and 23.2 AU, respectively. In this case the locations of secular resonances are shifted inward by about 9 AU compared with those in the present-day model. The distance between a KBO and Neptune is expanded and that between Jovian planets is compressed in this compact

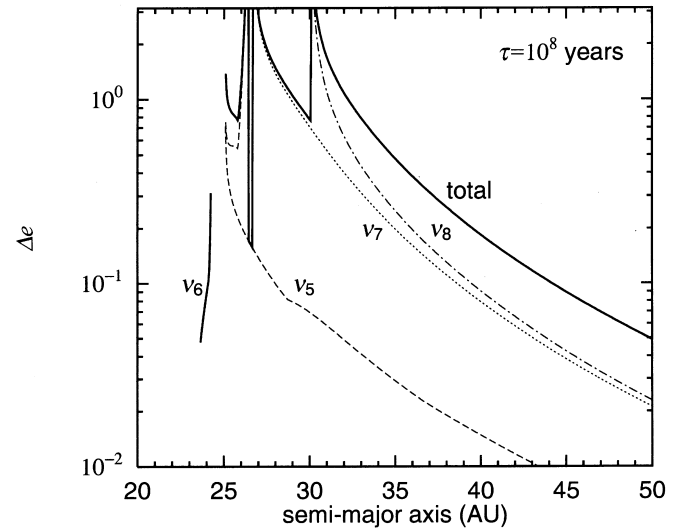


FIG. 16.—Estimated excitations in eccentricity in the compact model. Meanings of the lines are the same as those in Fig. 7, but the nebula depletion timescale $\tau = 10^8$ yr is adopted here. The secular resonances are shifted inward by about 9 AU compared with those in the present-day model.

model. The rotation of perihelion and ascending node of KBO is slower, while those of planets are faster. To make their rotation rates coincide with each other, KBOs need larger planetary perturbations, which are obtained near Neptune. As a result, the secular resonances occur at smaller semimajor axes than in the present planetary systems. To obtain e and i in the outer belt up to observed levels, $\tau \sim 10^8$ yr is required.

The initially compact planetary system should expand to the present configuration finally. When the nebula depletion timescale (T_{dep}) is shorter than the planetary migration timescale (T_{mig}), the results are similar to those in the compact model. When $T_{\text{dep}} > T_{\text{mig}}$, the results are similar to those in the present-day model. In the case of $T_{\text{dep}} \sim T_{\text{mig}}$, the results are not easily predicted. The results may be between the compact model and the present-day model, but

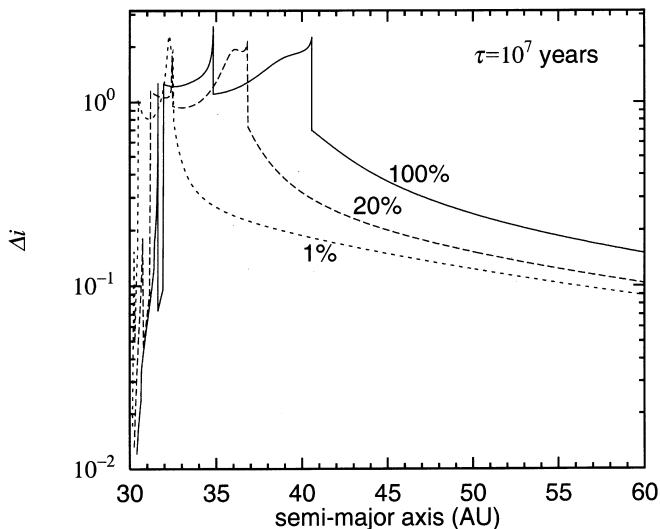


FIG. 15.—Estimated excitations in inclination in the small-mass model. The nebula depletion timescale is $\tau = 10^7$ yr. Meanings of lines are the same as those in Fig. 14.

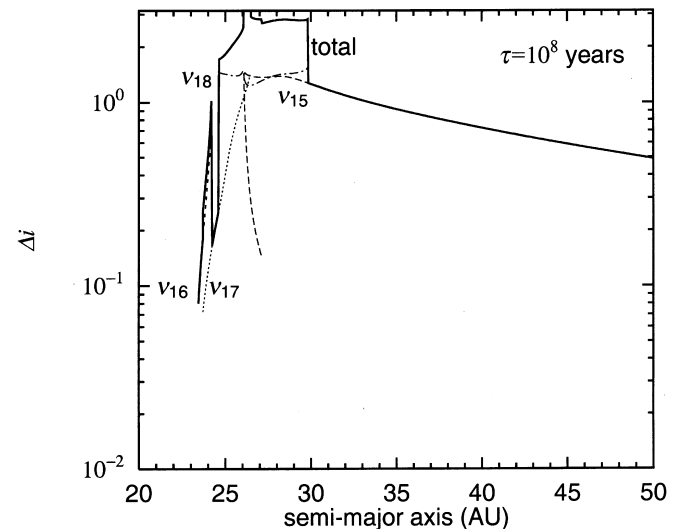


FIG. 17.—Estimated excitations in inclination in the compact model. Meanings of the lines are the same as those in Fig. 8. The nebula depletion timescale $\tau = 10^8$ yr is adopted.

more detailed calculation is needed. Note that T_{dep} is $|\rho(\partial\rho/\partial t)^{-1}|$ of a residual nebula with 0.1%–10% of initial density. We will discuss T_{dep} versus T_{mig} again in § 4.

4. CONCLUSIONS AND DISCUSSION

We have investigated orbital excitations of KBOs caused by the sweeping secular resonances during the primitive solar nebula depletion. The system we considered consists of the Sun, four Jovian planets, massless particles, and uniformly dissipating nebula. For the “starting” nebula model, the minimum mass solar nebula extending to 150 AU was basically employed. We also examined other nebula models and found that the results hardly change unless nebula surface density has nearly flat radial distribution. With the linear theory, we calculated paths of the secular resonance sweeping and excited eccentricities and inclinations of massless particles in the Kuiper belt region. We also performed direct orbital integrations in one case and found that the results of orbital integration agree with those of the linear theory. We found the following:

1. Inclinations (in radians) of KBOs are excited more than eccentricities.
2. Inclinations of KBOs are excited to the observational level by the sweeping secular resonances if the depletion timescale of the nebula $|\rho(\partial\rho/\partial t)^{-1}|$ is about 10^7 – 10^8 yr after the nebula density decreases to $\sim 10^{-3}$ of initial one. The existence of Uranus and Neptune are not necessarily required for the excitation of inclination, but in the cases without Uranus and Neptune, the required depletion timescale is about 10^8 yr.
3. Eccentricities of KBOs are excited to the observed level, if Neptune with more than one-fifth of the present mass exists and nebula depletion time is about 10^7 – 10^8 yr after nebula density becomes $\sim 10^{-1}$ of initial one.

The observed values of inclinations are larger than those of eccentricities in the outer belt. The eccentricities and inclinations of the outer KBOs are distributed up to $e \sim 0.2$ and $i \sim 0.6$, and root mean squares are $\langle e^2 \rangle^{1/2} \sim 0.1$ and $\langle i^2 \rangle^{1/2} \sim 0.2$. The sweeping of secular resonances explain

well large inclinations in the outer belt, while the other models (hypothetical Earth-sized planetesimals, hypothetical early stellar encounters, partial trapping by the migrating 2:1 resonance) more or less have difficulty to account for these large inclinations. For the excitations of eccentricities and inclinations to the observed maximum level ($e \sim 0.2$, $i \sim 0.6$), the required timescales of nebula depletion for the eccentricities and inclinations are consistent with each other. In Figure 18, maximum excitations of eccentricities and inclinations with various nebula depletion timescales (τ) are shown in the case of the present-day model. The excitation is proportional to $\sqrt{\tau}$ (eq. [5] and eq. [6]). Filled circles show the observed values of eccentricities and inclinations of KBOs in the outer belt. Since lines show the maximum values of excitation obtained by analytical estimation, observations do not necessarily reach these lines. Moreover, KBOs may interact with other KBOs; then the distribution of eccentricity and inclination may diffuse somewhat from analytical estimations. We find that the eccentricities and inclinations of KBOs are excited to the observational level if the nebula depletion timescale is 10^7 – 10^8 yr. (Since KBOs are unstable if $q \lesssim 35$ AU, the actual excitation of eccentricities can have been larger than presently observed eccentricities.)

The required nebula depletion timescale $\sim 10^7$ – 10^8 yr may be longer than that inferred from observations of T Tauri stars ($\sim 10^6$ – 10^7 yr) (Strom et al. 1993; Zuckerman et al. 1995). However, an observationally inferred timescale would correspond to a depletion timescale $|\rho(\partial\rho/\partial t)^{-1}|$ of a major fraction of the nebula, while our timescale refers to the depletion timescale of residual (0.1%–10% in mass) nebula gas. They are not necessarily the same.

We note that we neglected forced nebula motion caused by planets. Since planets excite density waves in the nebula gas, the structure of the nebula locally changes, and it may affect behavior of the secular resonances. We also neglected self-gravity of planetesimals disk. The disk particles show collective motion when they interact with each other, and Neptune launches density waves among them (Ward & Hahn 1998). Ward & Hahn (1998) show that this mechanism tends to inhibit the excitation of eccentricities owing

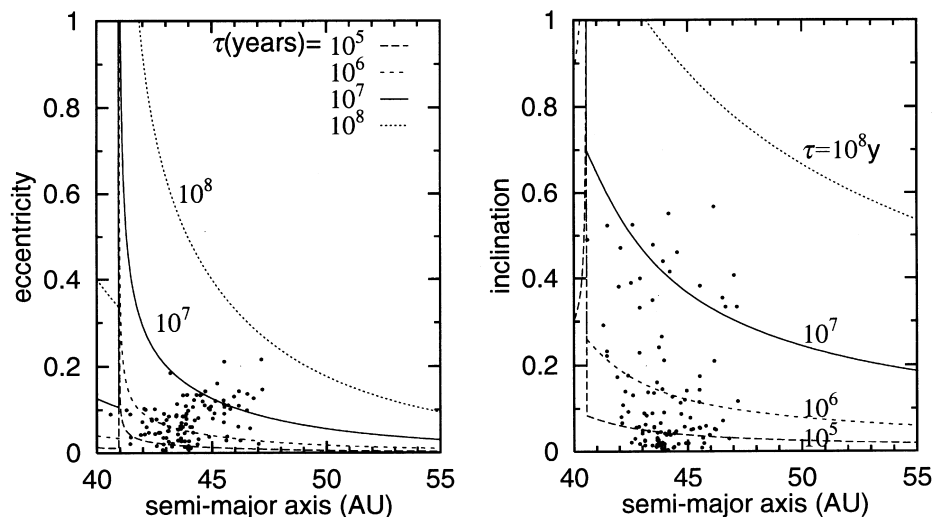


FIG. 18.—Nebula depletion timescale and observation of outer belt. Filled circles show the distribution of (left) eccentricities and (right) inclinations of the outer belt KBOs (see URL listed in footnote 1). Note that we removed the objects with pericenter distance is $\lesssim 35$ AU that would be considered to be the scattered disk objects. The lines show the maximum excitations of eccentricity and inclination with various nebula depletion timescales (τ) in the case of present-day model.

to secular resonances. As we showed, the secular resonances occur when the mass of nebula has become comparable to the mass of a planetesimal disk, so that depletion of the planetesimal disk may also be important. In the future papers, we will address these issues.

The observed concentration of KBOs near the location of the 3:2 mean motion resonance, however, cannot be explained by the sweeping secular resonance alone. We would need sweeping of the 3:2 mean motion resonance caused by planetary migration (Malhotra 1993, 1995). The timing of the mean motion resonance sweeping and the secular resonance one is delicate. The eccentricity's and inclination's secular resonances sweep the Kuiper belt when $u [= \rho(t)/\rho(0)]$ is 0.1 and when $u \sim 0.001$, respectively, that is, after depletion of the major fraction of the nebula. Hence, Neptune formation and hence the mean motion resonance sweeping due to Neptune's migration can precede the secular resonance sweeping. In this case, the present-day model is applied, and both concentration at the 3:2 resonance and excitation of e and i in the outer belt up to the observational level are realized if the residual nebula is depleted in 10^7 – 10^8 yr. On the other hand, if the secular resonance sweeping precedes the mean motion resonance sweeping, compact and/or small-mass model are applied, which is not favorable because of the longer depletion time-scale ($\gtrsim 10^8$ yr). Furthermore, eccentricities of the KBOs at $a < 39$ AU is pumped up before the mean motion resonance sweeping, which may inhibit effective capture by the 3:2 resonance (Malhotra 1995).

Several mechanisms have been proposed to explain e and i of the KBOs in the outer belt: Neptune-scattered planetesimals (Morbidelli & Valsecchi 1997; Petit et al. 1999), hypothetical stellar encounter (Ida et al. 2000), sweeping of 2:1 mean motion resonance (Hahn & Malhotra 1999), and sweeping secular resonances in the present paper. All of them may account for $\langle e^2 \rangle^{1/2} \sim 0.1$ in the outer belt. Although it would be difficult for the models of Neptune-scattered planetesimals and sweeping 2:1 mean motion

resonance to account for relatively large i , they cannot be ruled out completely at present. The hypothetical stellar encounter model may potentially account for it; however, we need more detailed examination.

We note that predicted e and i beyond 50 AU are quite different among these models. The model of hypothetical stellar encounter predicts that e and i rapidly increase beyond 50 AU since the effect of a passing star sensitively depends on the distance from the star (Ida et al. 2000). Thus this model predicts a “hot” disk beyond 50 AU and truncation of the disk farther out. On the other hand, the sweeping 2:1 mean motion resonance model predicts a very “cold” disk beyond 50 AU. There is not strong mean motion resonance between the 2:1 resonance (~ 48 AU) and the 3:1 resonance (~ 62 AU). The 3:1 resonance would be weaker than the 2:1 resonance for eccentricity excitation. e and i are hardly excited beyond 50 AU by this mechanism. The sweeping secular resonance and Neptune-scattered planetesimal models predict a moderately “cold” disk. In the case of the sweeping secular resonance, the magnitude of excited e and i systematically decreases with semimajor axis as shown in Figure 7 and Figure 8. The Neptune-scattered planetesimal model would not show a systematic trend of e and i , since excitations are caused by stochastic perturbations of the large planetesimals. But, the magnitude of e and i would generally decrease with semimajor axis. The number of KBOs so far detected beyond 50 AU is small. If future observation finds so many objects beyond 50 AU as to allow statistical discussion, it would evaluate the above models and consequently give important clues on what happened in the outer solar system in the formation stage.

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